NOVEMBER 2026 UPDATE
ANNA SOFIA MORO

SIMULATIONS

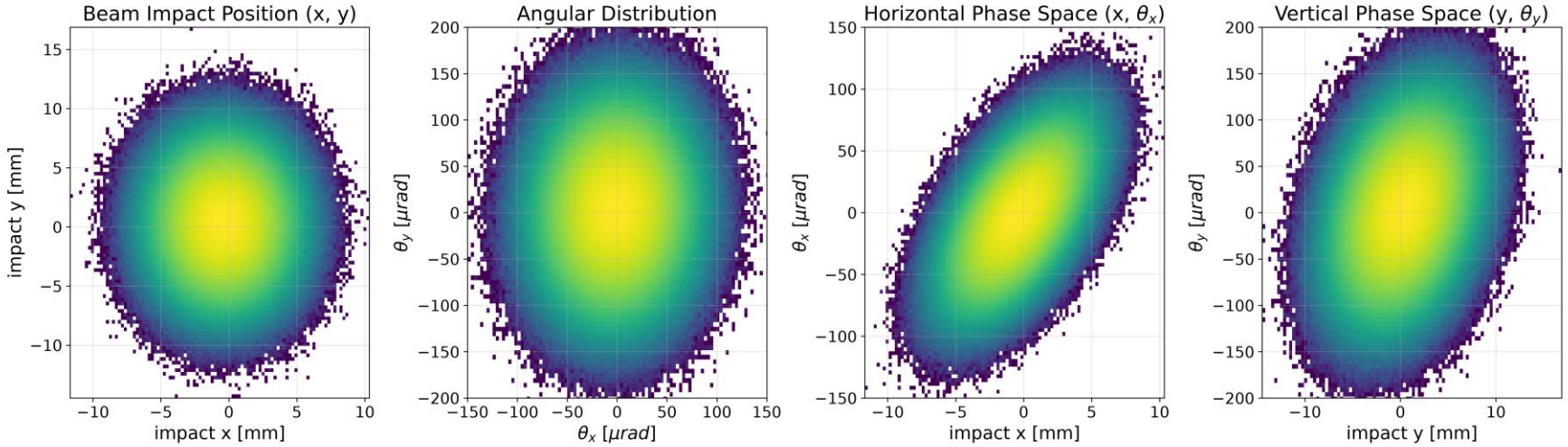
	BDSIM (GEANT4)	XCOLL (XSUITE)
SHORT CRYSTAL	1	2
LONG CRYSTAL	3	4

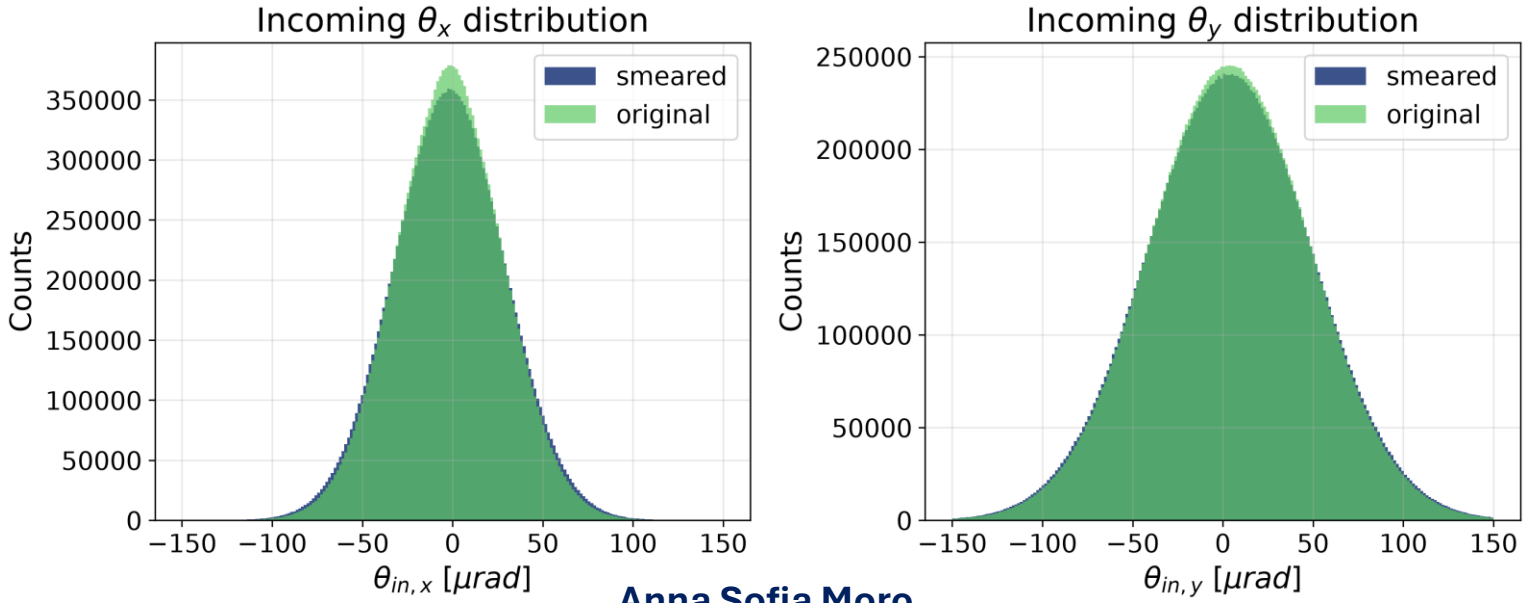
→ Compare crystal – different routines

→ Compare crystal – different routines **and data**

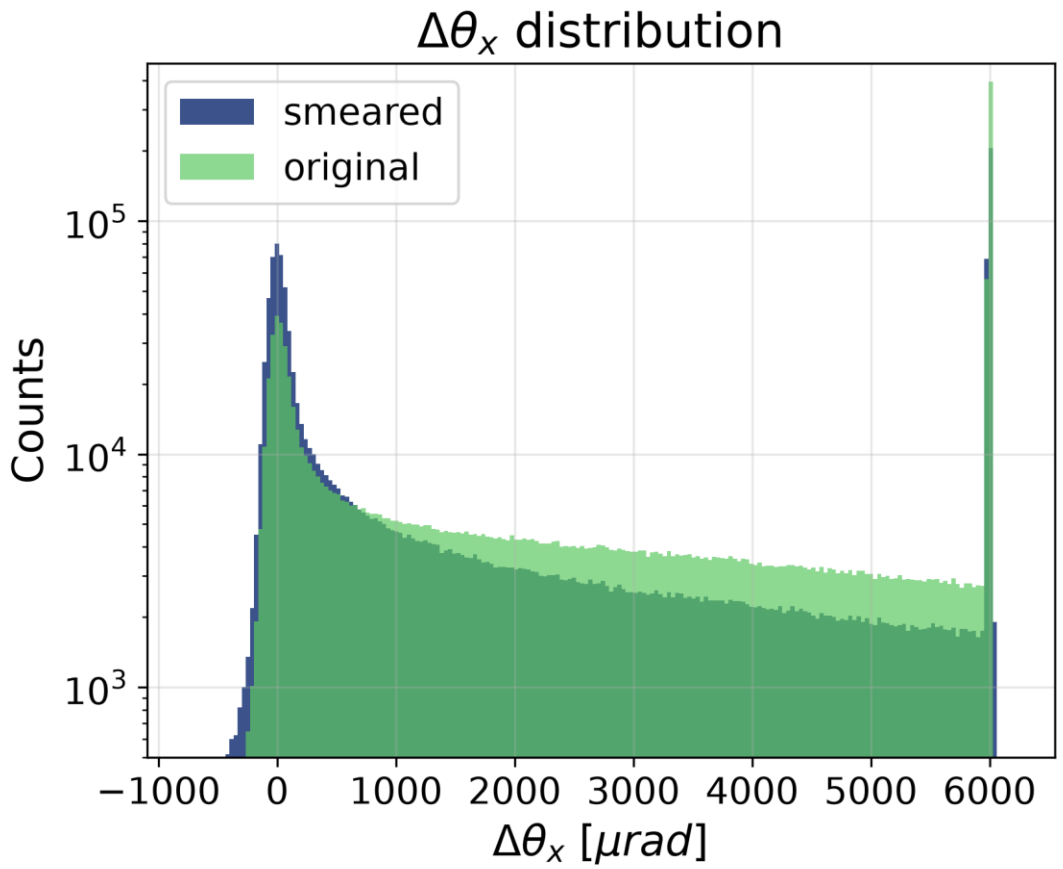
- Same beam distribution as real data
- Manually added detector resolution $\approx 9\ \mu\mathrm{rad}$
- Same analysis

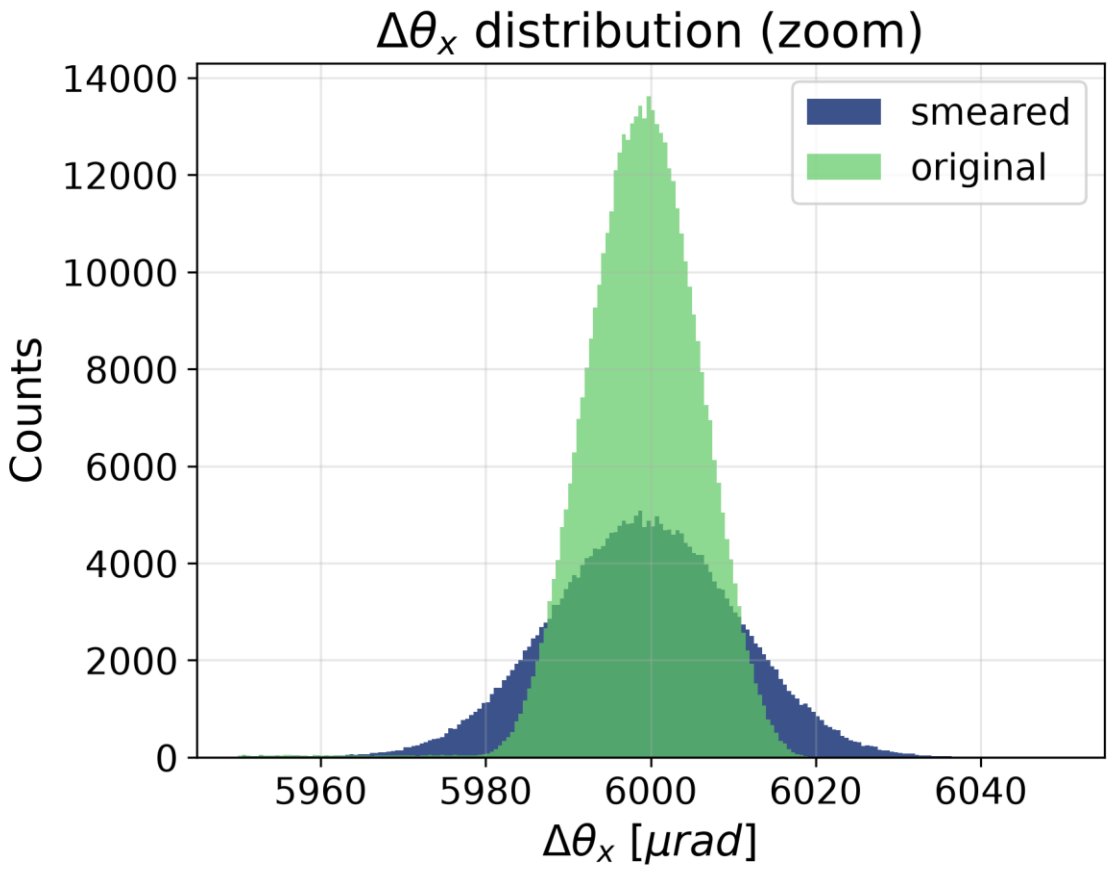
SIMULATIONS





3) BDSIM – LONG CRYSTAL

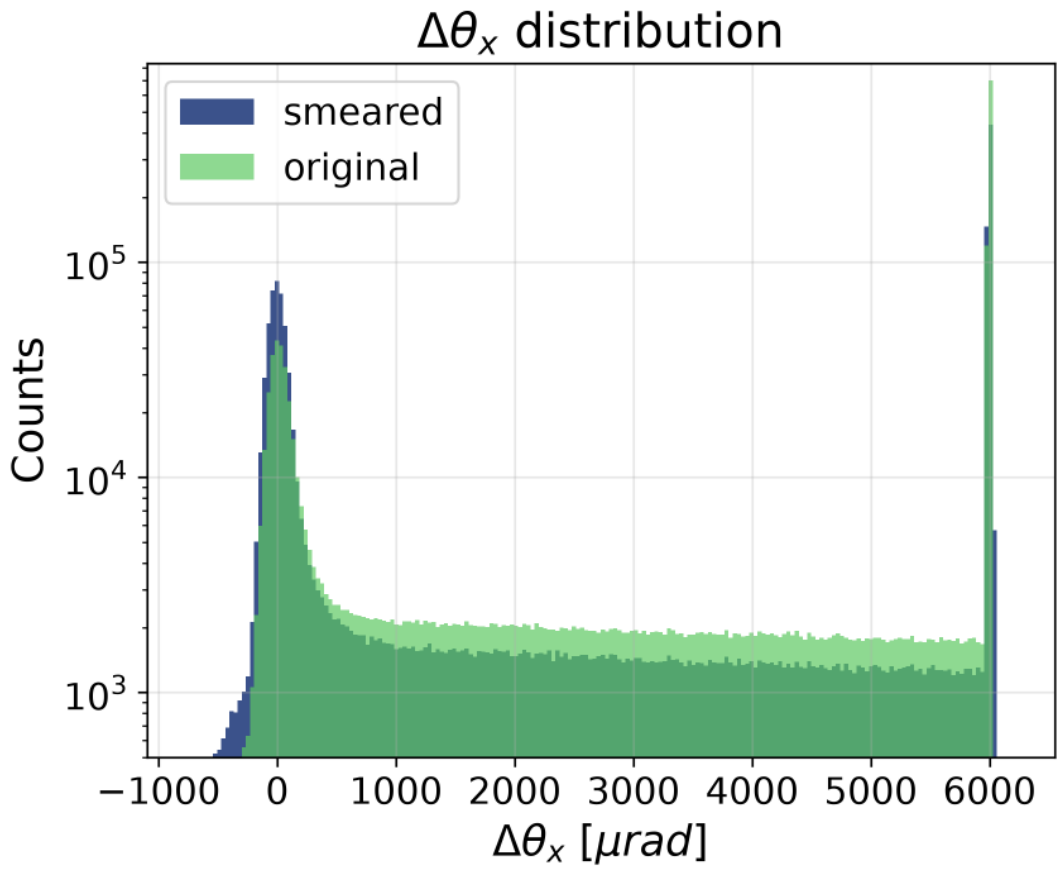


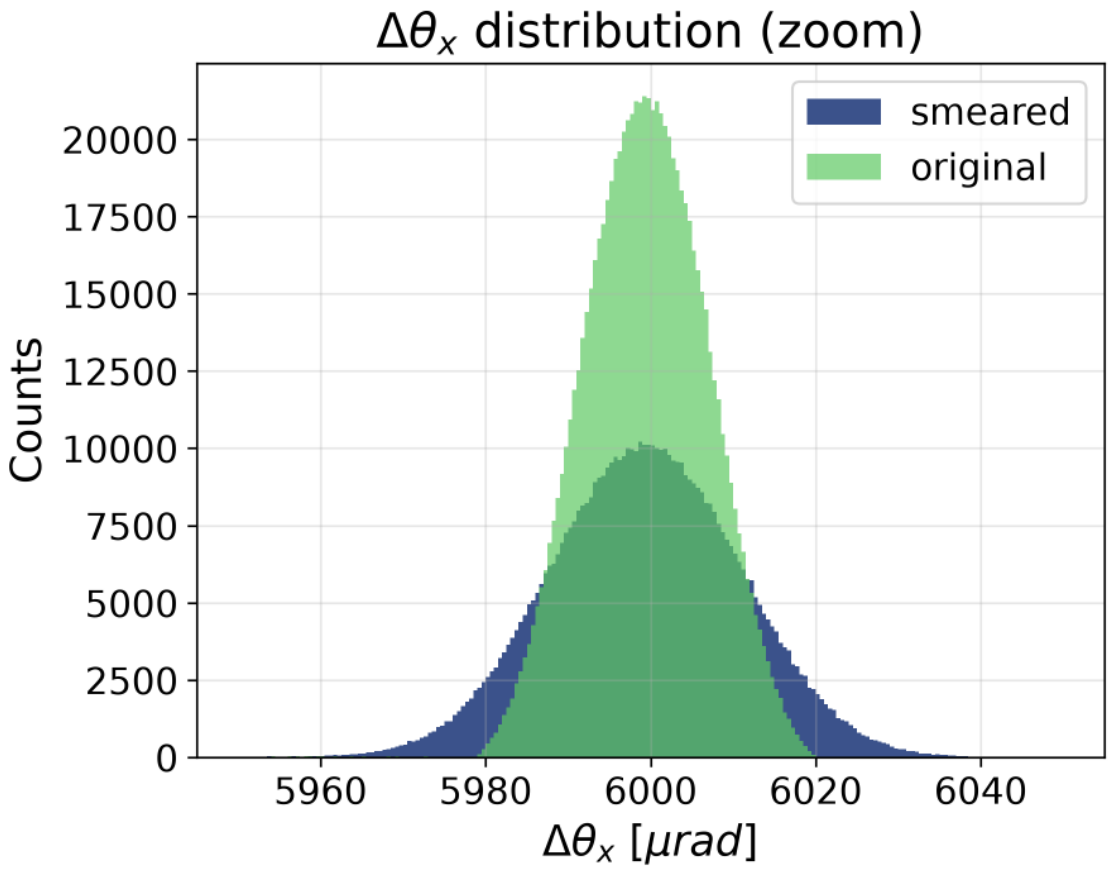


without smear $\varepsilon_{ch} = (30.3 \pm 0.6)\%$

with smear $\varepsilon_{ch} = (20.2 \pm 0.1)\%$

4) XCOLL – LONG CRYSTAL



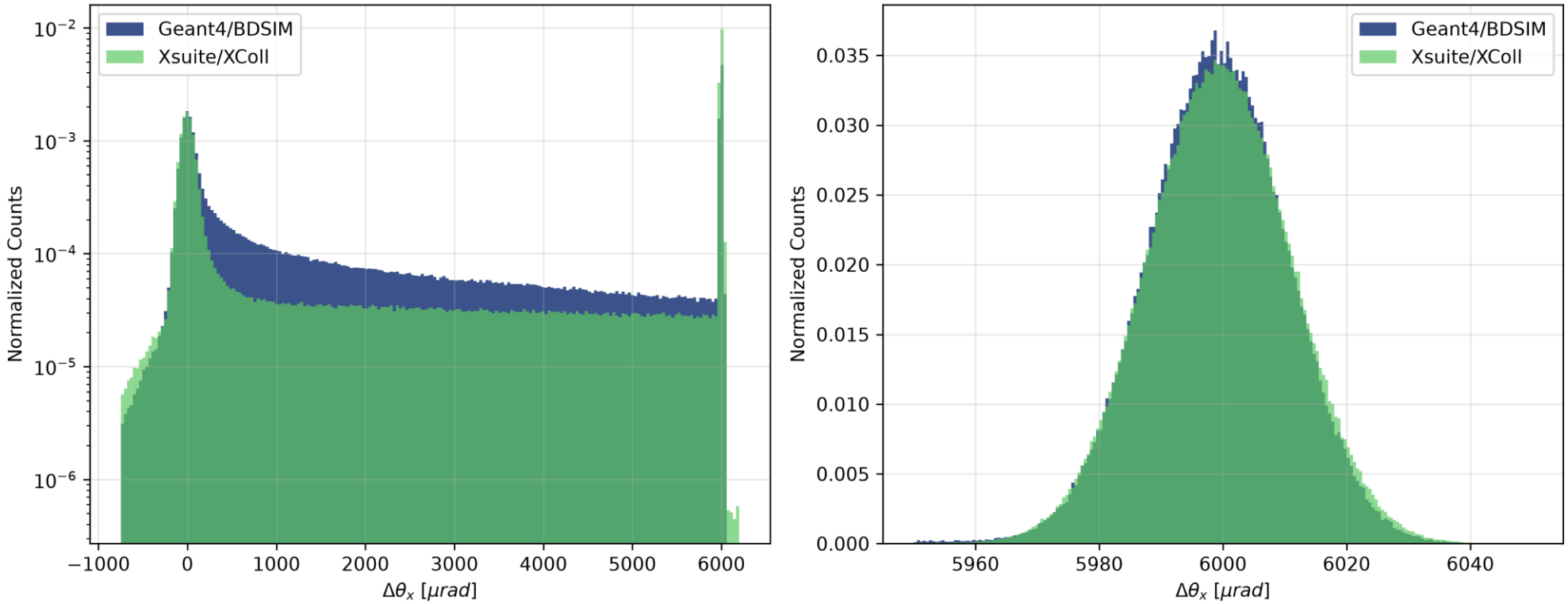


without smear $\varepsilon_{ch} = (49.0 \pm 0.9)\%$

with smear $\varepsilon_{ch} = (41.2 \pm 0.1)\%$

LONG CRYSTAL - COMPARISON

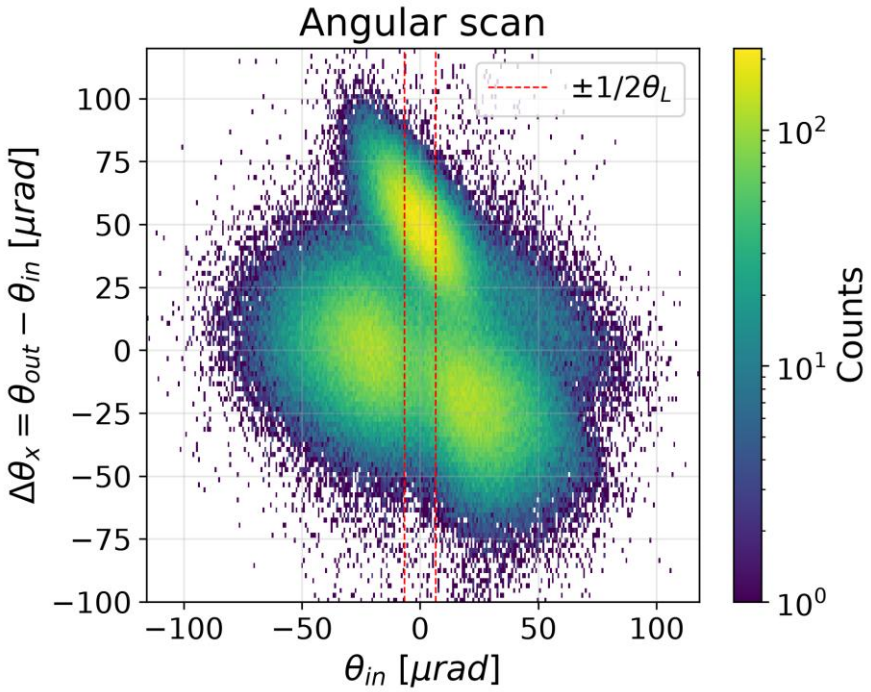
$\varepsilon_{ch} = (41.2 \pm 0.1)\%$ $\varepsilon_{ch} = (20.2 \pm 0.1)\%$

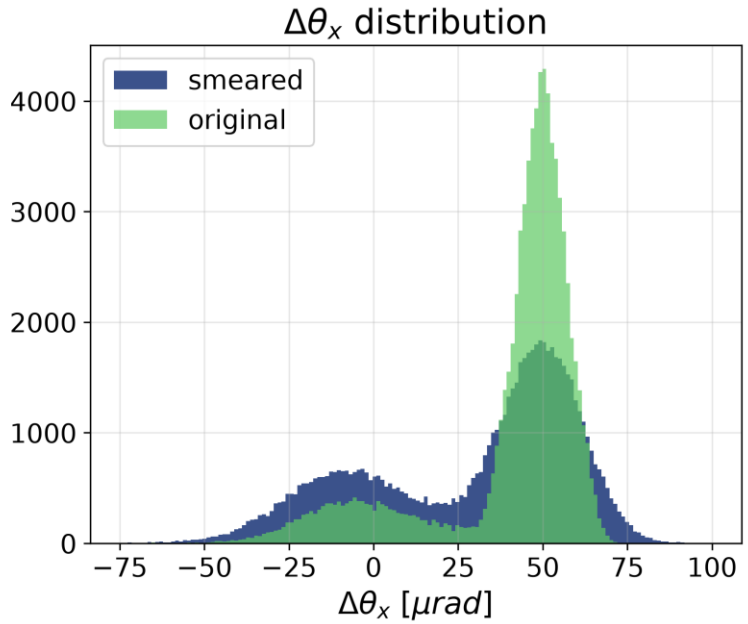


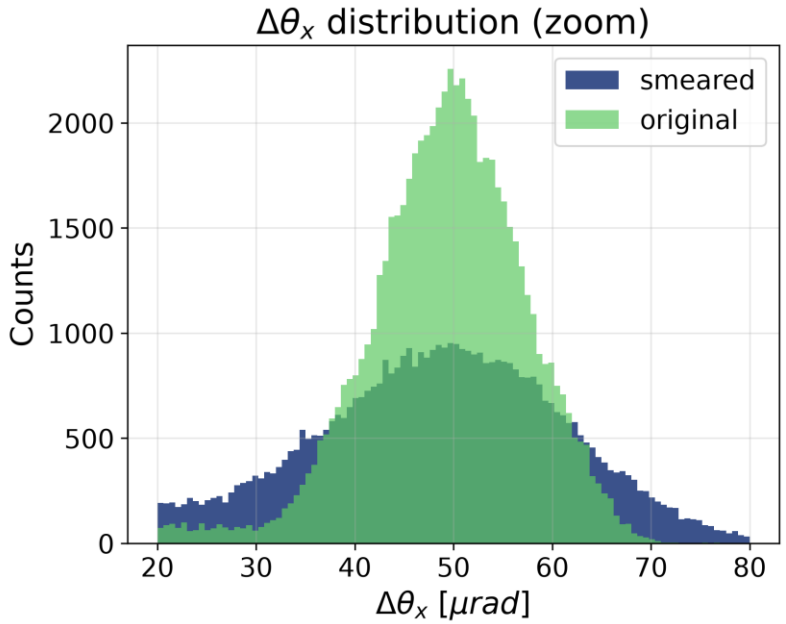
$\theta_b = 5999.5 \pm 0.1\ \mu\text{rad} \quad -\sigma = 11.5 \pm 0.1\ \mu\text{rad}$

$\theta_b = 5999.3 \pm 0.1\ \mu\text{rad} \quad -\sigma = 11.0 \pm 0.1\ \mu\text{rad}$

1) BDSIM – SHORT CRYSTAL



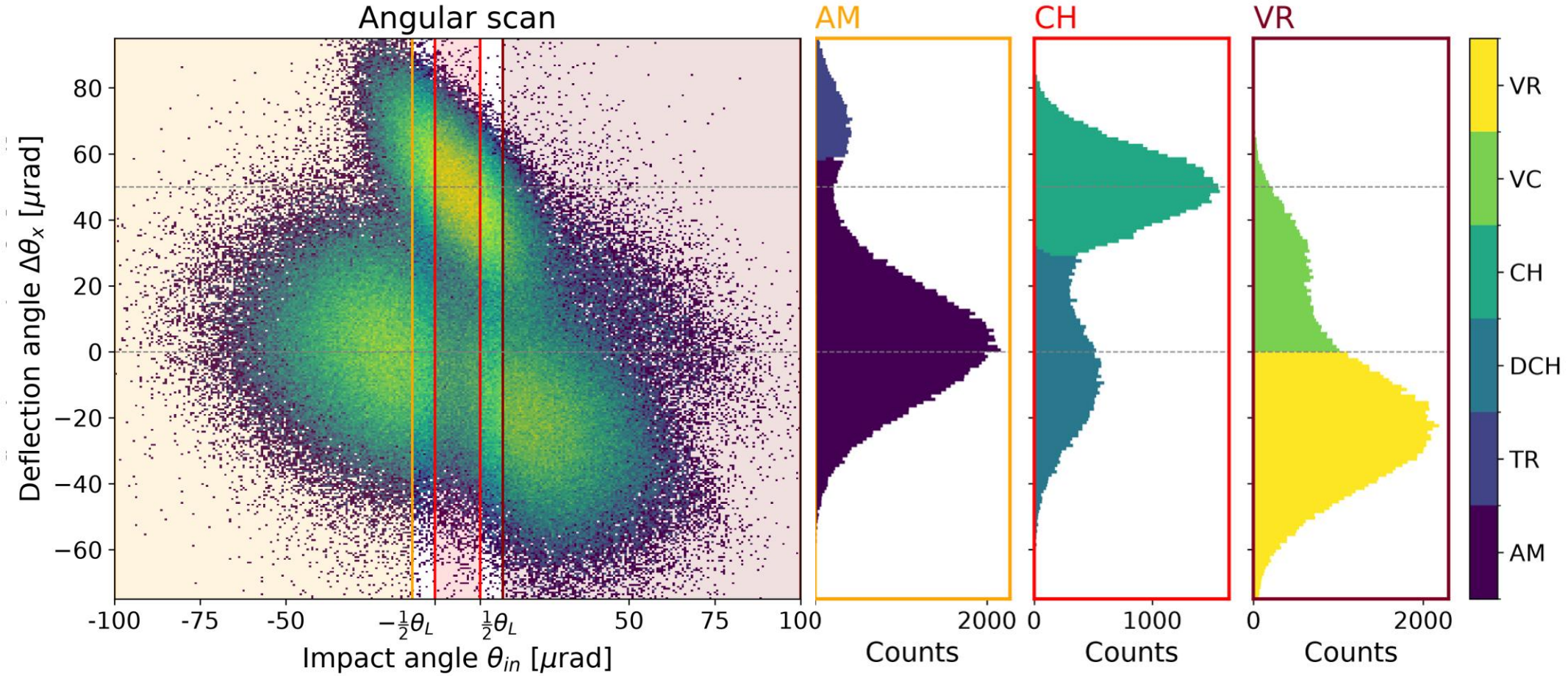




without smear $\boldsymbol{\varepsilon_{ch} = (80 \pm 1)\%}$

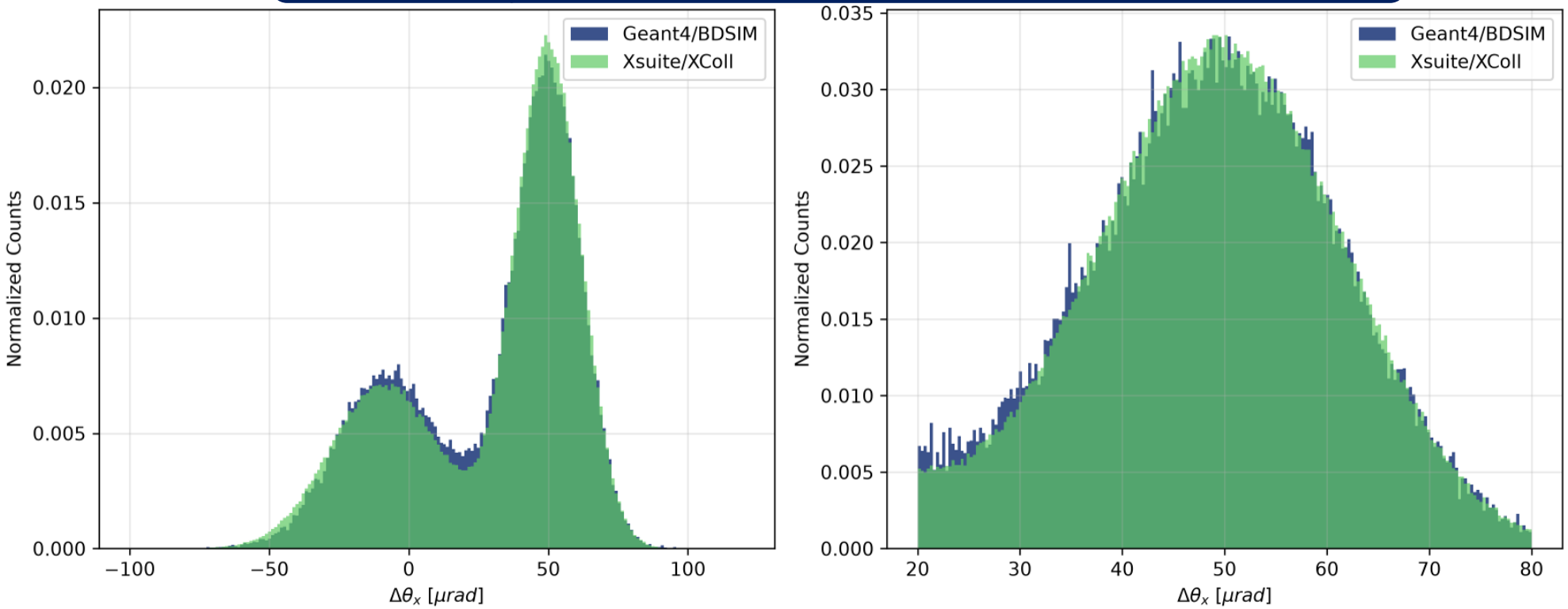
with smear $\boldsymbol{\varepsilon_{ch} = (62.8 \pm 0.6)\%}$

1) BDSIM – SHORT CRYSTAL



SHORT CRYSTAL - COMPARISON

$\varepsilon_{ch} = (62.8 \pm 0.6)\%$ $\varepsilon_{ch} = (60.1 \pm 0.3)\%$



$\theta_b = 49.9 \pm 0.2\ \mu\text{rad} \quad -\sigma = 11.8 \pm 0.1\ \mu\text{rad}$

$\theta_b = 50.1 \pm 0.2\ \mu\text{rad} \quad -\sigma = 11.9 \pm 0.1\ \mu\text{rad}$

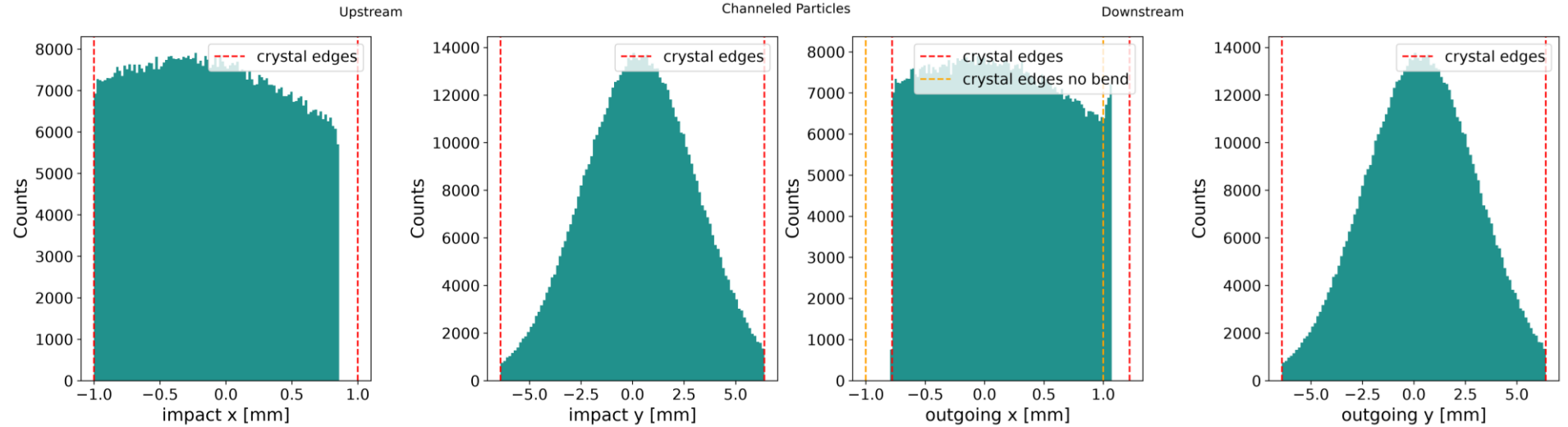
SIMULATIONS

	BDSIM (GEANT4)	XCOLL (XSUITE)	DATA
SHORT CRYSTAL	$\varepsilon_{ch} = (62.8 \pm 0.6)\%$	$\varepsilon_{ch} = (60.1 \pm 0.3)\%$	
LONG CRYSTAL	$\varepsilon_{ch} = (20.2 \pm 0.1)\%$	$\varepsilon_{ch} = (41.2 \pm 0.1)\%$	$\varepsilon_{ch} = (19.6 \pm 0.2)\%$

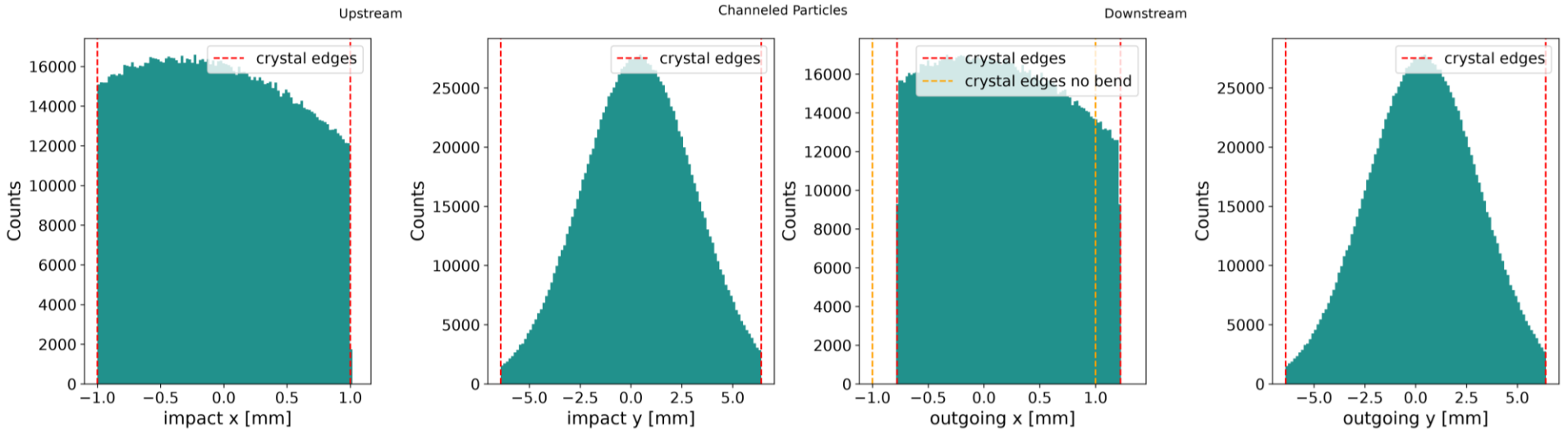
BACKUP

LONG CRYSTAL - COMPARISON

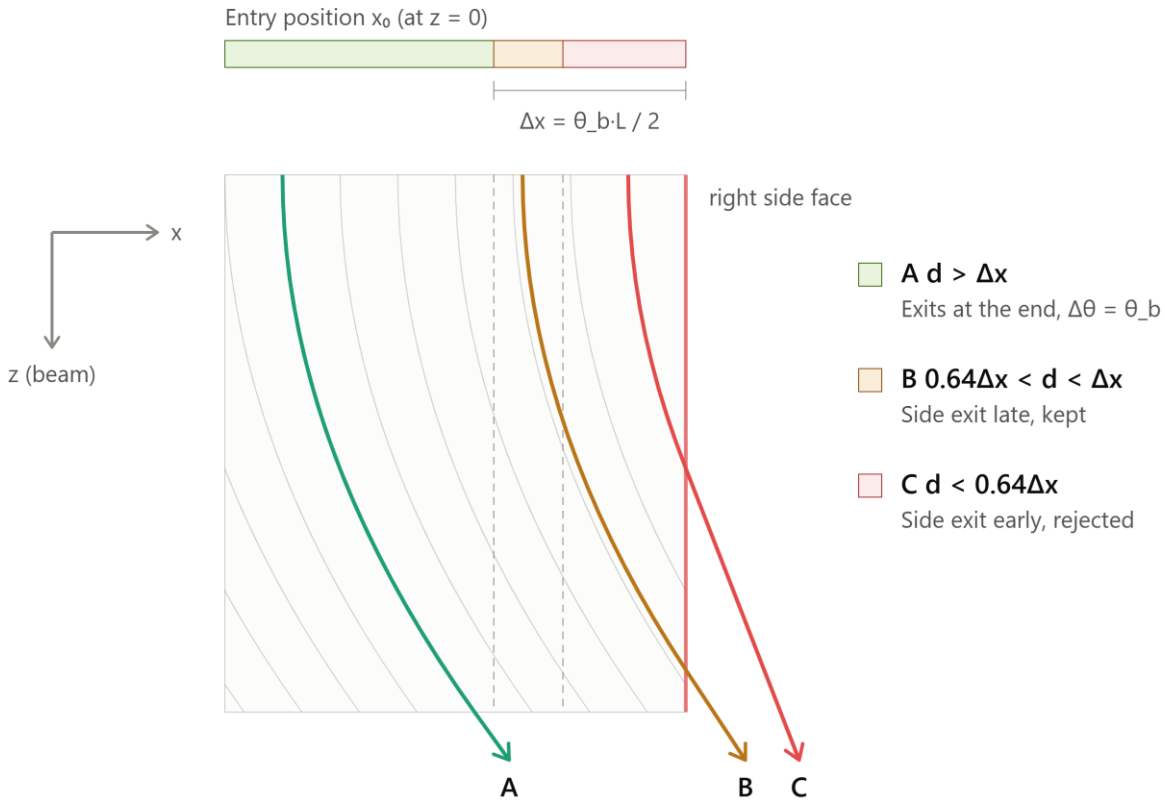
BDSIM

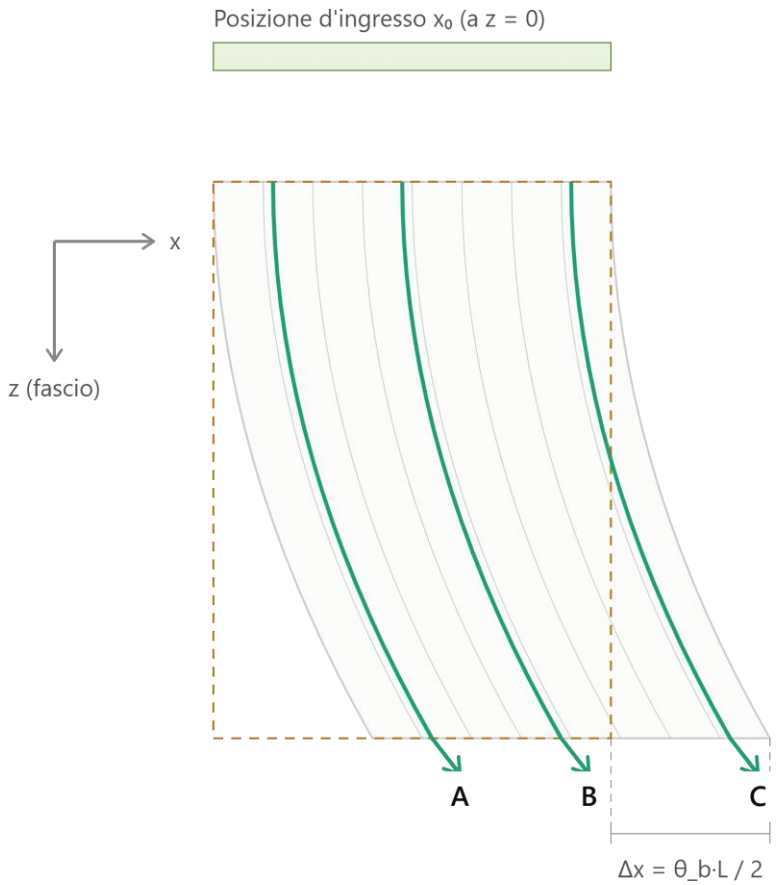


XCOLL

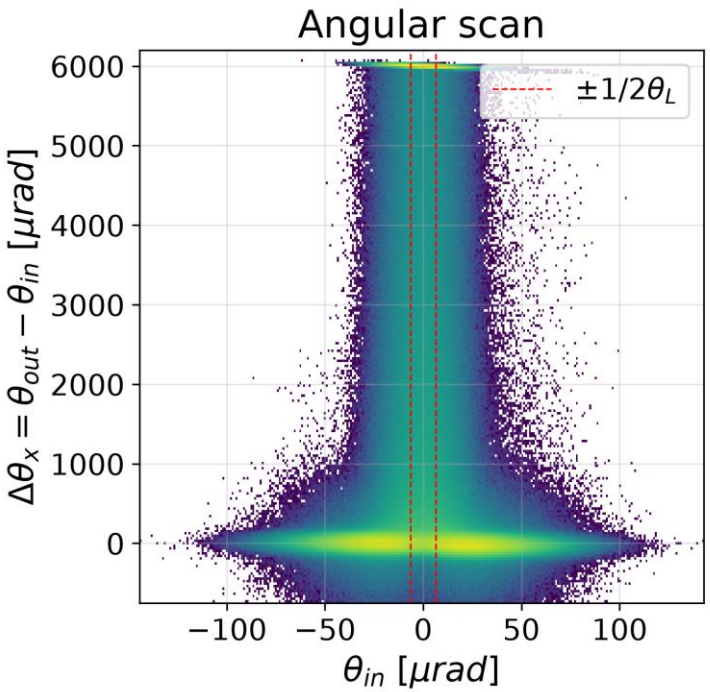


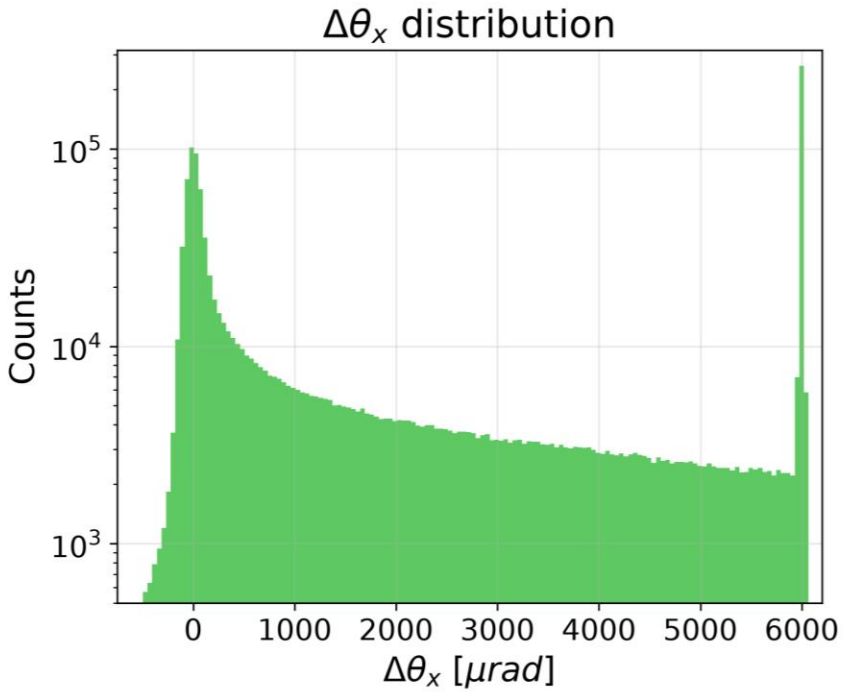
LONG CRYSTAL - COMPARISON

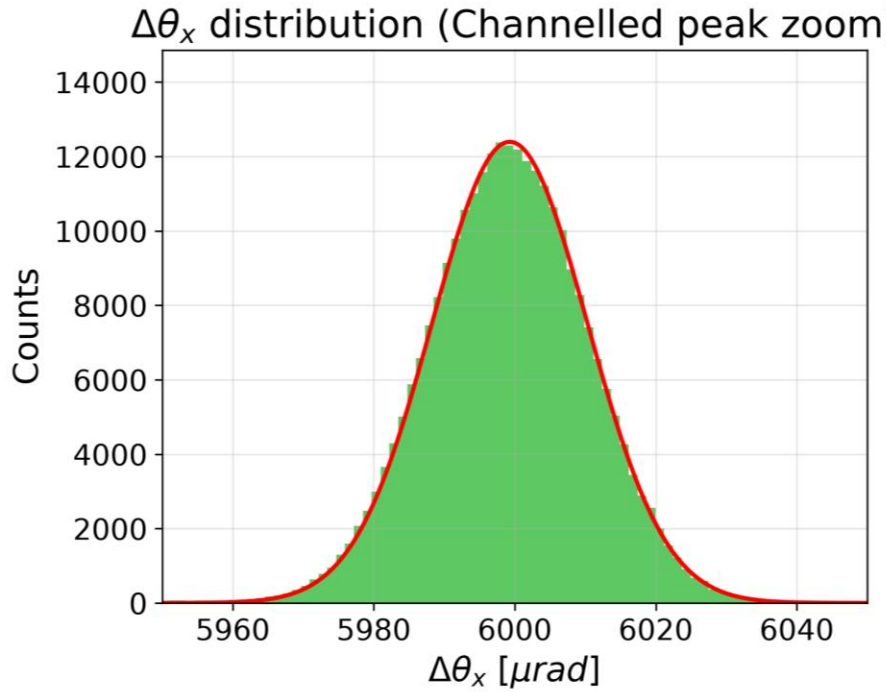




BDSIM – LONG CRYSTAL

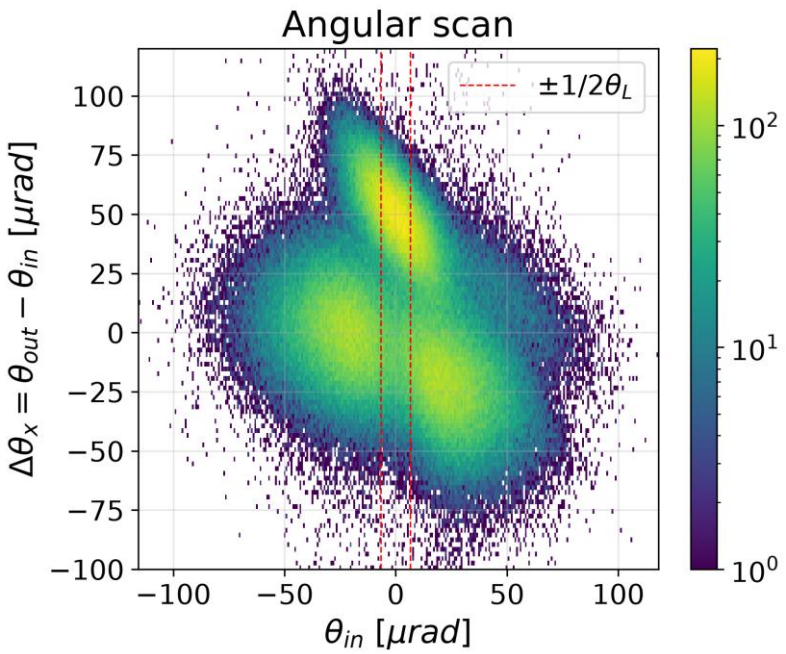


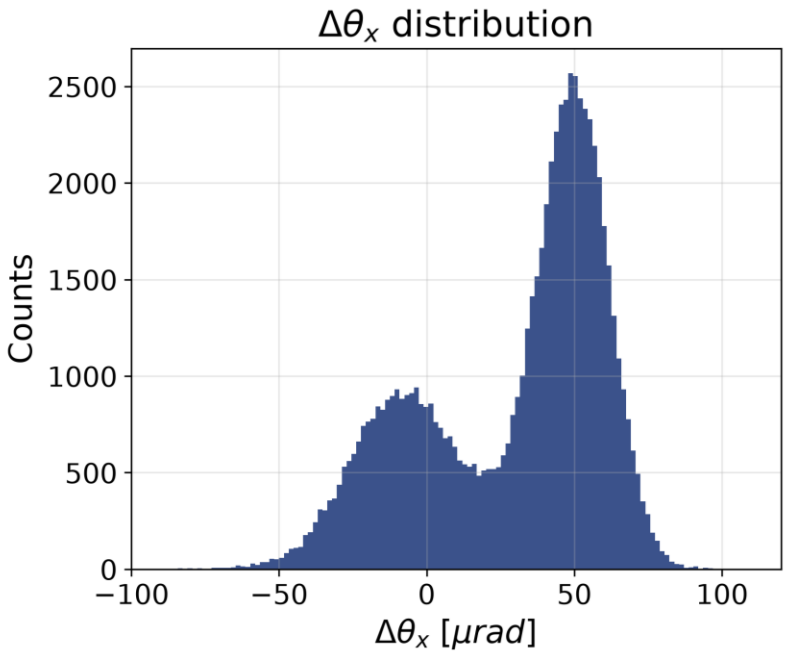


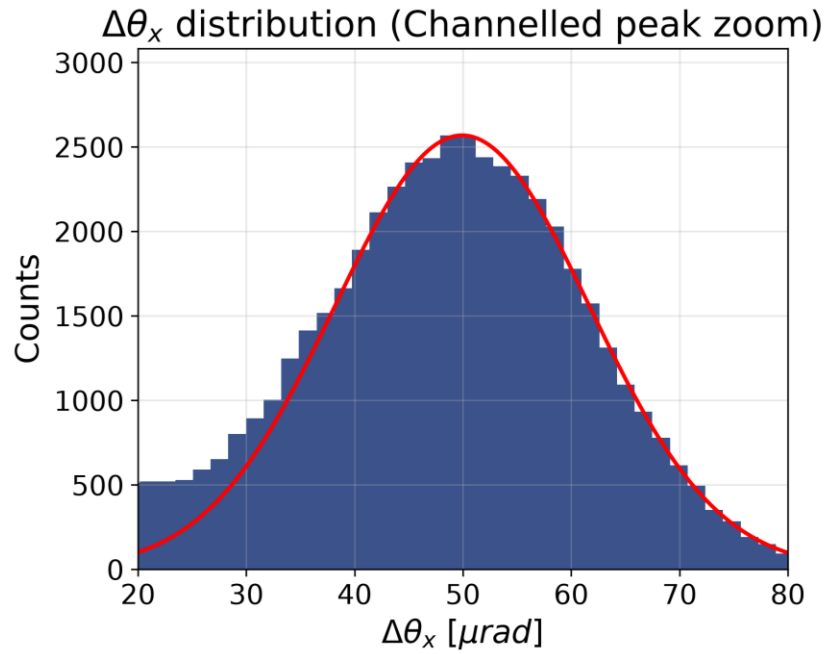


$$\theta_b = 5999.3 \pm 0.1\ \mu\text{rad}$$
$$\sigma = 11.0 \pm 0.1\ \mu\text{rad}$$

BDSIM – SHORT CRYSTAL







SHORT CRYSTAL

